MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIRST SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG1DSCCFS100 - INTRODUCTION TO DIGITAL FORENSICS AND TOOLS

Duration: 1.5 hrs MaximumMarks:50

Students should attempt at least one question from each course outcome to enhance their overall outcome attain ability.

Part A

Short Answer Type Questions
Answer **any 10** Questions
Each question carries **1** mark

1. Explain the use of digital forencics-intelligence. [Understand][1]
2. Identify the five basic stages of the digital forensic process. [Understand][1]
3. Identify the steps for collecting and preserving digital evidence [Understand][1,2]
4. Describe Chain of Custody. [Understand][2]
5. Identify any two hashing algorithms. [Understand][2]
6. Describe MRU. [Understand][1,2]
7. Summarize the concept of data sampling. [Understand][1,3]
8. Outline the concept of obfuscation. [Analyse][1,3]
9. Examine which tool helps investigators to analyse hard drives,smartphones for digital evidence [Analyse][3]
10. What is FTK imager? [Understand][1,4]
11. Name the image analysis tool. [Apply][4]
12. List out some tools used for data masking. [Apply][4]

[10x1=10]

Part B

Short Essay Type Questions
Answer **any 4** Questions
Each question carries **5** marks

13. Identify the uses of digital forensics. [Understand][1]
14. Describe the process in the identification and preservation of digital evidence. [Understand][1,2]
15. Explain Order of Volatility. [Understand][2]
16. Explain Hybrid sleep. [Understand] [1,2]
17. Differentiate NTFS and FAT. [Analyse][1,3]
18. How can we carve data using FDAC? [Apply[4]

[4x5=20]

Part C

Essay Type Questions

Answer **any 2** Questions

Each question carries **10** marks

19. Describe the five basic stages of the digital forensic process and their importance in a legal investigation. [Understand][1]
20. Describe in detail, about evidence acquisition from a live system. [Understand][2]
21. Discuss the various types of password attacks, their methodologies, and the potential impact on individuals and organizations. Compare effective strategies for mitigating these attacks and enhancing password security. [Understand][1,3]

[2x10=20]